https://xtremepape.rs/

Question	Answer	Marks
1	**Defining the problem**	
	d is the independent variable and *R* is the dependent variable **or** vary *d* and measure *R*	**1**
	keep intensity/power of light source constant	**1**
	Methods of data collection	
	labelled diagram showing a light source fixed above container of water with the labelled LDR positioned in the beaker	**1**
	correct circuit diagram to measure *R*, e.g. *V* and *I* methods or ohmmeter	**1**
	method to determine *R*, e.g. $R = \frac{\text{p.d. across LDR}}{\text{current}}$ or read off ohmmeter	**1**
	method to determine *d*, e.g. use a ruler or drawn labelled vertical ruler adjacent to container with *d* indicated	**1**
	Method of analysis	
	plots a graph of *R* against d^2	**1**
	relationship valid if a straight line produced passing through the origin	**1**
	$K = \frac{4\pi}{\text{gradient}}$	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**Max. 6**
	D1 dark glasses to prevent damage to eyes due to light source **or** do not look directly at light source **or** do not touch hot lamp/use gloves to position hot lamp/heat-proof gloves to position lamp	
	D2 dark room or shielding LDR (so as to avoid light from other sources)	
	D3 use high intensity lamp or collimated beam or laser	
	D4 method described to check that current in light source is constant, e.g. use an ammeter and variable resistor / variable power supply	
	D5 keep position of light source constant or distance between light source and LDR constant	
	D6 light source is placed close to water surface to increase intensity/reduce reflections **or** light source is placed further away to make it more directional	
	D7 use tall container to give a wide range of d or R or to reduce uncertainties **or** use a wide container to reduce reflections	
	D8 method to position ruler vertically to measure d described e.g. use a set square/spirit level	
	D9 use of horizontal fiducial mark from ruler to meniscus or middle of LDR, e.g. pin **or** d = reading on rule at surface – reading at top of LDR	
	D10 ensure that the electrical connections/wire to the LDR are waterproof	

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